04

Lecture 04

Mobility and Handover

تحرک و تحویل

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 1–3; log-distance path loss; Gaussian distribution and the Q-function

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-04-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Distinguish** idle-mode mobility (cell reselection, location registration) from connected-mode mobility (handover) and explain who decides in each case.
- **LO2** **Compare** location areas, routing areas, tracking-area lists and RAN notification areas, and **optimise** the size of a registration area with a cost model.
- **LO3** **Classify** handovers (intra-/inter-frequency, inter-RAT, hard/soft) and state the measurement quantities RSRP and RSRQ.
- **LO4** **Apply** the LTE A3 entering condition with hysteresis, offsets and time-to-trigger, and **compute** where a handover is triggered.
- **LO5** **Trace** an X2-based LTE handover message by message, including data forwarding and path switch.
- **LO6** **Diagnose** too-late, too-early, wrong-cell and ping-pong handovers and relate them to parameter settings.

2 Prerequisite knowledge

You need the location-area concept of Lectures 1–2, the soft handover of WCDMA (Lecture 3), and the log-distance path-loss model with log-normal shadowing: $L_{\mathrm{p}}(d) = L_{\mathrm{p}}(d_0) + 10n\log_{10}(d/d_0) + X_\sigma$, where X_σ is a zero-mean Gaussian variable (in dB) with standard deviation σ. The tail function is $Q(x) = \Pr\{Z > x\}$ for a standard Gaussian Z.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Mobility management	مدیریت تحرک	Keeping a moving user reachable and connected.
Cell reselection	انتخاب مجدد سلول	UE-controlled change of camping cell in idle mode.
Handover (HO)	تحویل	Network-controlled transfer of an active connection to another cell.
Tracking area (TA)	ناحیهٔ ردیابی	LTE/5G registration area; the paging scope.
RSRP	توان دریافتی سیگنال مرجع	Average power of one cell-specific reference-signal resource element.
RSRQ	کیفیت دریافتی سیگنال مرجع	$N \cdot \mathrm{RSRP}/\mathrm{RSSI}$: a load- and interference-aware quality.
Hysteresis	پسماند	Margin that prevents repeated triggering around a threshold.
Time-to-trigger (TTT)	زمان تا راه‌اندازی	Duration a condition must hold before a report is sent.
Ping-pong	پینگ‌پنگ	Unnecessary handover followed quickly by a return.
Radio link failure (RLF)	شکست پیوند رادیویی	Loss of the connection detected by timers (e.g. T310).

3 Introduction and motivation

A fixed telephone is always at the same socket; a mobile user is not. Mobility management answers two different questions:

(a) **Reachability** – when the user is idle, where should the network look for it when a call or a packet arrives?

(b) **Session continuity** – when the user is active, how can the connection follow the user from cell to cell without dropping or interrupting it?

Both questions involve a trade-off between *signalling cost* and *performance*. If the network knew the exact cell of every idle user, every cell change would cost an update; if it knew nothing, every incoming call would have to be paged everywhere. If handovers are triggered eagerly, users bounce between cells (ping-pong); if they are triggered late, links fail. This lecture develops both sides across 2G to 5G and gives each trade-off a quantitative form.

4 Fundamental concepts

4.1 Idle mode versus connected mode

In **idle mode** the UE has no radio connection. It *camps* on a cell, listens to its broadcast and paging channels, and chooses by itself when to move to a better cell (cell reselection). The network only knows the UE's registration area. In **connected mode** the UE has a dedicated radio connection; the network decides the serving cell, based on measurement reports from the UE (network-controlled, mobile-assisted handover).

4.2 Registration areas across generations

- **Location area (LA)** – GSM/UMTS CS domain; updated towards the VLR.
- **Routing area (RA)** – GPRS/UMTS PS domain; a subset of an LA, updated towards the SGSN. Two parallel procedures were needed because the two domains were separate.
- **Tracking area (TA)** – LTE and 5G. The MME (AMF in 5G) gives the UE a *list* of TAs; the UE updates only when it enters a TA outside the list. Lists can be tailored per UE, which removes the ping-pong of updates at a fixed border.
- **RAN notification area (RNA)** – 5G RRC_INACTIVE state: the core considers the UE connected, the RAN keeps its context and pages it within a RAN-configured area.

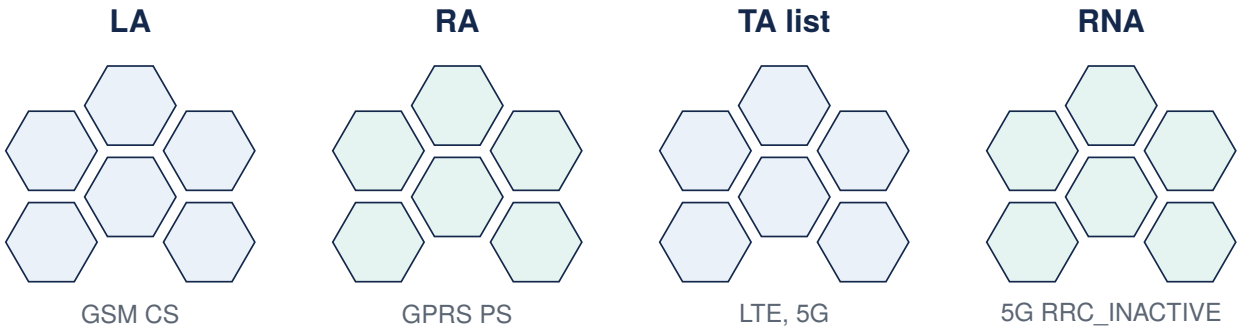


Figure 1. Registration areas of successive generations.

4.3 Measurement quantities

LTE defines two basic measurements on the cell-specific reference signals [1]:

Definition 1 • RSRP and RSRQ

RSRP is the linear average of the power of the resource elements that carry cell-specific reference signals within the measured bandwidth (range about -140 to -44 dBm). *RSRQ* is

$$\mathrm{RSRQ} = \frac{N \cdot \mathrm{RSRP}}{\mathrm{RSSI}}, \tag{1}$$

where N is the number of resource blocks in the RSSI measurement bandwidth and RSSI is the total received power (wanted signal, interference and noise) over that bandwidth.

RSRP measures coverage; RSRQ also reflects load and interference, because RSSI grows when neighbours are busy. NR uses the analogous SS-RSRP and SS-RSRQ on synchronisation signal

blocks.

5 Detailed technical explanation

5.1 Cell reselection in idle mode

In LTE a cell is *suitable* if it satisfies the S-criterion [3]:

$$S_{\text{rxlev}} = Q_{\text{rxlevmeas}} - (Q_{\text{rxlevmin}} + Q_{\text{rxlevminoffset}}) - P_{\text{compensation}} > 0, \tag{2}$$

where $Q_{\text{rxlevmeas}}$ is the measured RSRP, Q_{rxlevmin} the broadcast minimum level and $P_{\text{compensation}}$ a penalty for UEs whose maximum power is below the level assumed by the operator. Among suitable intra-frequency cells the UE ranks the serving and neighbour cells with

$$R_{\text{s}} = Q_{\text{meas,s}} + Q_{\text{hyst}}, \qquad R_{\text{n}} = Q_{\text{meas,n}} - Q_{\text{offset}}, \tag{3}$$

and reselects a neighbour only if it is ranked best for a time $T_{\text{reselection}}$. Inter-frequency and inter-RAT reselection use absolute *priorities*: a UE always moves to a higher-priority layer that is good enough, and leaves to a lower-priority layer only when its current layer becomes poor. Priorities let operators steer idle users towards, for example, a capacity layer or a 5G carrier.

5.2 Handover classification

- By **frequency**: *intra-frequency* (same carrier; measurable without gaps), *inter-frequency* (another carrier; the UE needs measurement gaps unless it has a second receiver) and *inter-RAT* (another technology, e.g. LTE to UMTS).
- By **connection continuity**: *hard* handover (break-before-make; GSM, LTE, NR) and *soft* handover (make-before-break; WCDMA, where several cells transmit the same data simultaneously).
- By **network scope**: within one controller (intra-BSC, intra-eNB), between base stations (X2/Xn handover) or involving the core (inter-MSC in GSM, S1-based in LTE, N2-based in 5G).

Handover in GSM. The BTS measures the uplink and the MS reports the six strongest neighbours every 480 ms on the SACCH. The BSC decides; inter-BSC handovers involve the MSC, and inter-MSC handovers keep the first MSC as an *anchor* for the call [4].

Soft handover in WCDMA. The UE maintains an *active set* of cells; events add a cell that comes within a reporting range of the best cell, remove a cell that falls out of it, or replace the weakest member. The cost is extra downlink power and Iub capacity, typically 30–40% of users being in soft handover.

Handover in LTE. LTE removed the RNC and soft handover. The eNB decides alone, based on measurement *events* configured through RRC [2]:

5.3 Handover in 5G NR

NR keeps the LTE principles (eNB→gNB, X2→Xn, S1→N2) and adds features aimed at the interruption and the failures of hard handover [5]:

Table 1. LTE intra-RAT measurement events (thresholds and offsets are configured by RRC).

Event	Entering condition (simplified)	Typical use
A1	Serving becomes better than a threshold	Stop inter-frequency measurements (remove gaps)
A2	Serving becomes worse than a threshold	Start inter-frequency/inter-RAT measurements
A3	Neighbour becomes offset better than serving (PCell)	Intra-frequency handover
A4	Neighbour becomes better than a threshold	Load-based or coverage-layer handover
A5	Serving worse than threshold 1 and neighbour better than threshold 2	Inter-frequency handover
A6	Neighbour becomes offset better than a secondary cell	Carrier-aggregation management
B1/B2	Inter-RAT neighbour better than a threshold (and serving worse, B2)	LTE to UMTS/GSM, LTE to NR

- **Conditional handover (CHO)**: the network prepares one or more target cells in advance and sends the UE a handover command with an execution condition; the UE executes it when the condition is met, avoiding the race in which the command arrives over a failing link.
- **Dual active protocol stack (DAPS)**: the UE keeps receiving from the source while connecting to the target, approaching zero interruption.
- **Beam management** inside a cell: many beam changes are handled at layers 1/2 without RRC handover.

5.4 Handover failures

A UE declares *radio link failure* (RLF) when, after $N310$ consecutive out-of-sync indications, timer $T310$ expires without recovery; it then attempts RRC re-establishment. Self-organising networks classify failures to tune parameters (mobility robustness optimisation, MRO):

- **Too late** – RLF in the source cell before (or during) the handover; the UE re-establishes in the target. Remedy: earlier triggering (lower offset/TTT).
- **Too early** – RLF shortly after a successful handover; the UE re-establishes in the *source*. Remedy: later triggering.
- **Wrong cell** – RLF shortly after handover; re-establishment in a third cell. Remedy: adjust the cell-individual offsets of the two neighbours.
- **Ping-pong** – a successful handover followed by a return to the source within a short time; wastes signalling and causes interruptions. Remedy: more hysteresis or TTT.

Too-late and ping-pong pull the parameters in opposite directions: this is the central tuning dilemma of handover.

6 Mathematical models

6.1 The A3 condition and the handover point

Engineering problem. Where along a road between two cells does a handover actually happen, and how do the parameters move that point?

The A3 entering condition. In LTE [2] the condition, with all quantities in dB or dBm, is

$$M_{\mathrm{n}} + O_{\mathrm{fn}} + O_{\mathrm{cn}} - \mathrm{Hys} \;>\; M_{\mathrm{p}} + O_{\mathrm{fp}} + O_{\mathrm{cp}} + \mathrm{Off}, \tag{4}$$

where M_{n}, M_{p} are the (layer-3 filtered) measurements of the neighbour and the primary serving cell, $O_{\mathrm{f\cdot}}$ frequency-specific offsets, $O_{\mathrm{c\cdot}}$ cell-individual offsets, Hys the hysteresis and Off the event offset. The condition must hold continuously for T_{TTT} before the UE reports. With zero frequency and cell offsets it reduces to $M_{\mathrm{n}} - M_{\mathrm{p}} > H$ with $H = \mathrm{Off} + \mathrm{Hys}$.

System model. Two eNBs at distance D with equal transmit power; the UE moves along the line joining them; deterministic path loss with exponent n (no shadowing). At distance x from the serving eNB,

$$M_{\mathrm{n}} - M_{\mathrm{p}} = 10n \log_{10} \frac{x}{D - x}. \tag{5}$$

Derivation. Setting $10n \log_{10}\big(x/(D-x)\big) = H$ and solving,

$$x_{\mathrm{HO}} = D\,\frac{r}{1+r}, \qquad r = 10^{H/(10n)}. \tag{6}$$

The UE then travels a further $v\,T_{\mathrm{TTT}}$ before reporting.

Interpretation. For $D = 1000\,\mathrm{m}$, $n = 3.5$ and $H = 3\,\mathrm{dB}$: $r = 1.218$ and $x_{\mathrm{HO}} = 549\,\mathrm{m}$, i.e. 49 m past the cell border; $H = 6\,\mathrm{dB}$ moves it to 597 m. At 30 m/s a TTT of 320 ms adds another 9.6 m (Figure 2). Every decibel of margin is paid for with signal quality on the old cell, where the link is already the weakest – the origin of too-late failures at high speed.

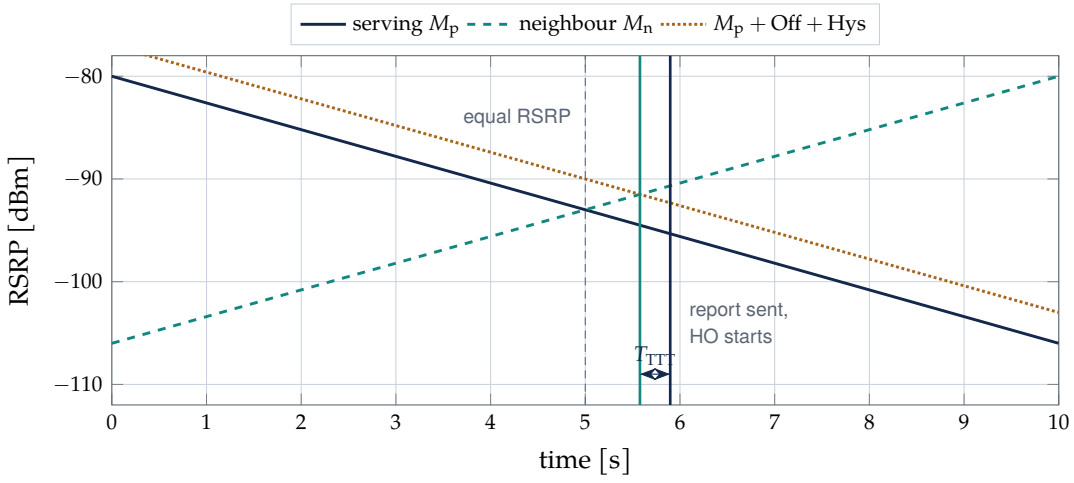


Figure 2. A3 triggering: the neighbour must exceed the serving cell by $\mathrm{Off} + \mathrm{Hys} = 3\,\mathrm{dB}$ (dotted) for $T_{\mathrm{TTT}} = 320\,\mathrm{ms}$ before the measurement report is sent.

6.2 Shadowing and ping-pong

Model. Add independent log-normal shadowing $X_{\mathrm{s}}, X_{\mathrm{n}} \sim \mathcal{N}(0, \sigma^2)$ (dB) to both cells. At the cell border the mean difference is zero, and $M_{\mathrm{n}} - M_{\mathrm{p}} \sim \mathcal{N}(0, 2\sigma^2)$.

Result. The probability that a single sample at the border exceeds the margin H is

$$P_{\text{trig}} = Q\left(\frac{H}{\sqrt{2}\,\sigma}\right). \tag{7}$$

With $\sigma = 8$ dB and $H = 3$ dB, $P_{\text{trig}} = Q(0.265) = 0.40$; even with $H = 6$ dB it is still 0.30. Hysteresis alone cannot suppress spurious triggers caused by large-scale fading; TTT and layer-3 filtering are needed. The layer-3 filter

$$F_k = (1-a)\,F_{k-1} + a\,M_k, \qquad a = 2^{-K/4}, \tag{8}$$

with filter coefficient K, reduces the variance of *independent* fluctuations (fast fading, measurement error) by the factor $a/(2-a)$ – one third for $K = 4$ – but it cannot remove shadowing, which is correlated over tens of metres. The model is pessimistic in one respect (it ignores the correlation of shadowing between the two cells) and optimistic in another (it looks at one point only); simulation along trajectories is the usual next step.

6.3 Sizing a registration area

Problem. How many cells should a location or tracking area contain?

Model and assumptions. Square cells of side a; a registration area of N cells forms a square of side $a\sqrt{N}$ and perimeter $L = 4a\sqrt{N}$. Users are uniformly distributed and move in uniformly random directions at mean speed v (the *fluid-flow* model [6]): the rate at which users of an area S cross its perimeter L is $\rho v L/\pi$, where ρ is the user density. Each user receives λ_c calls per unit time, each paged in all N cells. One update costs c_u and paging one cell costs c_p.

Derivation. Per user, the update rate is $\rho v L/(\pi\rho S) = vL/(\pi S)$ with $S = Na^2$, so

$$C(N) = c_u\,\frac{4v}{\pi a\sqrt{N}} + c_p\,\lambda_c\,N. \tag{9}$$

The first term decreases and the second increases with N. Setting $\mathrm{d}C/\mathrm{d}N = 0$,

$$-\frac{2c_u v}{\pi a}N^{-3/2} + c_p\lambda_c = 0 \quad\Longrightarrow\quad N^{*} = \left(\frac{2\,c_u\,v}{\pi\,a\,c_p\,\lambda_c}\right)^{2/3}. \tag{10}$$

Interpretation. Fast users (v large) favour large areas; users who receive many calls (λ_c large) favour small ones. Because the optimum depends on the user, LTE's per-UE tracking-area lists can approach it user by user, which a fixed LA plan cannot.

7 Network architecture and diagrams

The LTE handover architecture is flat: the source and target eNBs negotiate directly over X2, and the MME and S-GW are involved only at the end to move the downlink path (*path switch*). If no X2 exists, or if the MME or S-GW must change, the same steps run through the MME over S1 (*S1-based handover*). In 5G the roles map to gNB, Xn, AMF/SMF and UPF (Lecture 20). GSM and UMTS involve their controllers (BSC/RNC) and, across controller borders, the MSC/SGSN.

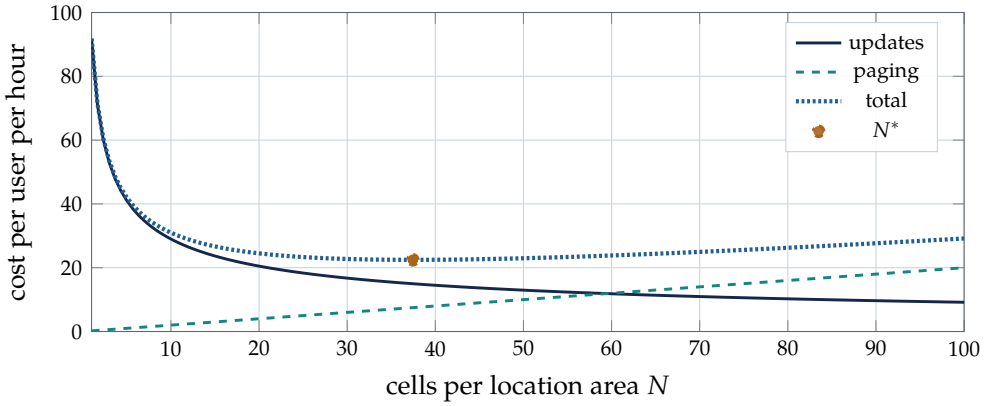


Figure 3. Update, paging and total signalling cost of Eq. (9) for $c_u = 1$, $c_p = 0.1$, $v = 10\,\mathrm{m/s}$, $a = 500\,\mathrm{m}$, $\lambda_c = 2$ calls per hour (Example 3).

8 Step-by-step procedures

Procedure 1 • LTE X2-based handover

1. **Preparation trigger.** The UE sends a Measurement Report after the A3 condition held for TTT. The source eNB decides to hand over.
2. **Preparation.** The source sends X2AP Handover Request with the UE context (bearers, security, RRC configuration). The target performs admission control, reserves resources (including a dedicated RACH preamble) and answers Handover Request Acknowledge containing the RRC message for the UE.
3. **Execution.** The source sends RRC Connection Reconfiguration with mobility control information, stops scheduling the UE, sends SN Status Transfer and starts forwarding buffered and in-transit downlink packets to the target over X2.
4. **Access.** The UE synchronises to the target, performs contention-free random access (Lecture 8), receives a timing advance and an uplink grant, and sends RRC Connection Reconfiguration Complete. The interruption ends.
5. **Completion.** The target asks the MME to move the path with S1AP Path Switch Request; the MME sends Modify Bearer Request to the S-GW, which switches the downlink tunnel and sends *end marker* packets on the old path so that the target knows when forwarded data is finished.
6. **Release.** After Path Switch Request Acknowledge, the target sends UE Context Release; the source frees its resources.

The user-plane interruption lasts from the moment the UE leaves the source until it can be scheduled in the target; it is of the order of a few tens of milliseconds in LTE and is the quantity that RACH-less handover, make-before-break and DAPS reduce.

9 Worked examples

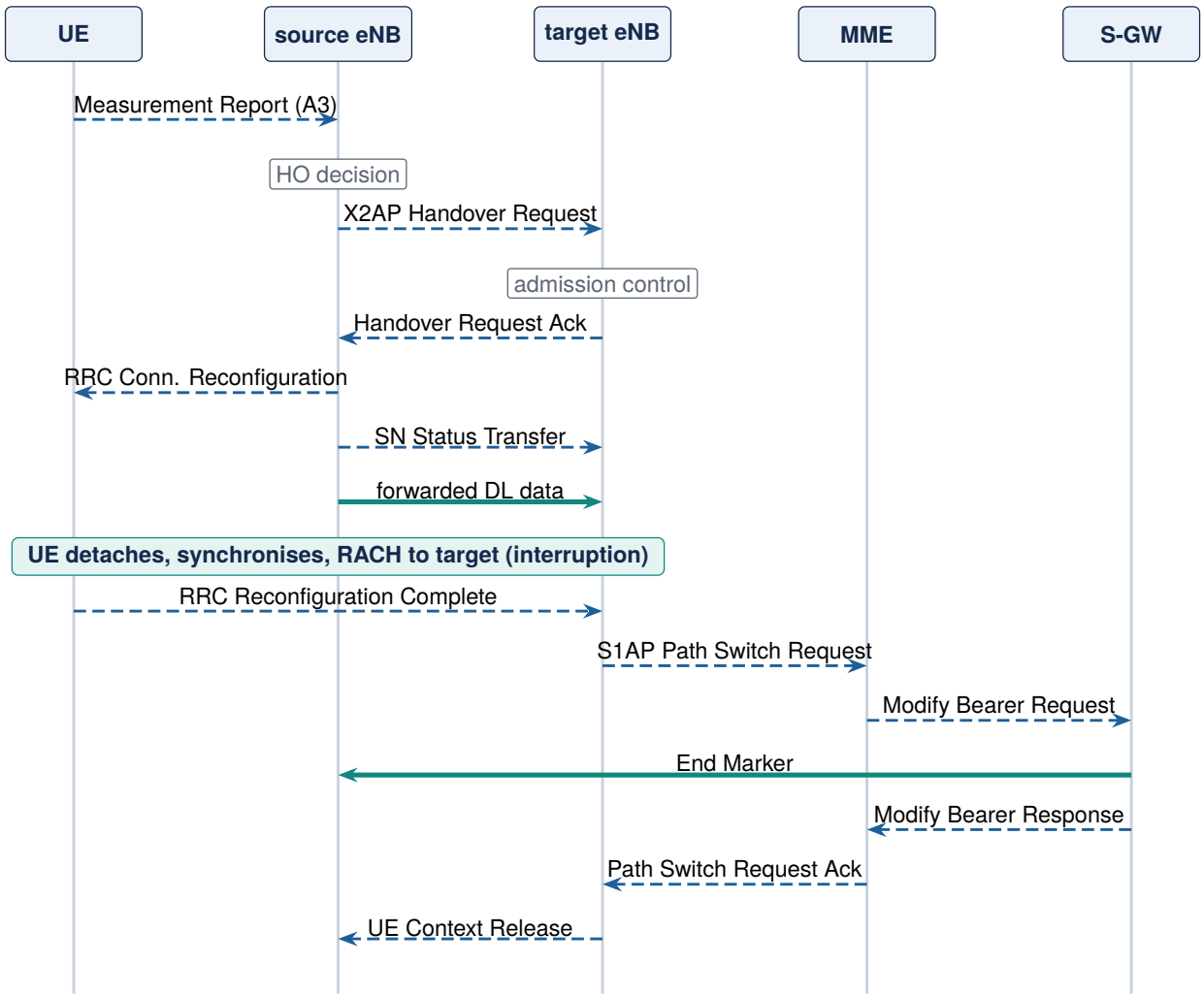


Figure 4. X2-based handover in LTE (Procedure 1). Teal arrows: user-plane data forwarding and end marker.

Worked example 1 • Handover point and speed

Problem. Inter-site distance 1000 m, $n = 3.5$, Off + Hys = 3 dB, $T_{\mathrm{TTT}} = 320$ ms. Where is the measurement report sent for a pedestrian (1.5 m/s) and for a car on a motorway (120 km/h)?
Solution. From Eq. (6), $x_{\mathrm{HO}} = 549$ m. The pedestrian adds $1.5 \times 0.32 = 0.5$ m; the car adds $33.3 \times 0.32 = 10.7$ m, reporting at ≈ 560 m.
Interpretation. Then preparation and execution take further tens of milliseconds, during which the car moves another 1–3 m per 30 ms. Motorway cells are therefore often given a lower offset or TTT, and speed-dependent scaling of TTT is part of the standard.

Worked example 2 • RSRQ under load

Problem. A UE measures RSRP $= -90$ dBm over $N = 50$ resource blocks. Compute RSRQ when the RSSI is -65 dBm (lightly loaded neighbours) and -60 dBm (heavily loaded).
Solution. RSRQ $= 10 \log_{10} 50 +$ RSRP $-$ RSSI $= 17.0 - 90 + 65 = -8.0$ dB, and $17.0 - 90 + 60 = -13.0$ dB.
Interpretation. Same coverage, 5 dB worse quality: RSRQ sees the load of the neighbours, RSRP does not. Using RSRQ for A3 spreads load, at the cost of more volatile reports.

Worked example 3 • Tracking-area size

Problem. With the parameters of Figure 3, find the optimal number of cells per area and the cost per user and hour.
Solution. $\lambda_c = 2/3600\,\mathrm{s}^{-1}$. Equation (10): $N^* = \left(2 \times 1 \times 10/(\pi \times 500 \times 0.1 \times 2/3600)\right)^{2/3} = (229.2)^{2/3} = 37.5$. A 6×6 area ($N = 36$) gives $C = 3600\,[4 \times 10/(\pi \times 500 \times 6) + 0.1 \times (2/3600) \times 36] = 15.28 + 7.20 = 22.5$ cost units per user-hour, against 91.9 for $N = 1$ and 29.2 for $N = 100$.
Interpretation. The optimum is flat: anything between about 20 and 60 cells costs within 10% of the minimum. Operators can therefore choose area borders for practical reasons (along low-traffic lines, away from busy roads) without losing much.

10 Technology comparison

Hard versus soft. Soft handover gives diversity gain and no interruption, but needs same-frequency neighbours, a central combining point (the RNC) and extra resources. LTE chose a flat architecture without a central RAN node, so soft handover was impossible; it compensated with fast hard handover, data forwarding and, later, DAPS. The trade-off – *centralised combining versus flat, fast decisions* – reappears in 5G with the CU/DU split (Lecture 14).

11 Key takeaways

Table 2. Connected-mode mobility from 2G to 5G.

	GSM	UMTS (WCDMA)	LTE	NR (5G)
Type	Hard	Soft (same freq.) + hard	Hard	Hard; CHO and DAPS options
Decision	BSC	RNC	Source eNB	Source gNB
Measurements	RXLEV, RXQUAL	CPICH RSCP, E_c/N_0	RSRP, RSRQ, SINR	SS-RSRP, SS-RSRQ, SS-SINR, CSI-RS
Inter-node link	A (via MSC)	Iur	X2 (or S1)	Xn (or N2)
Core involvement	MSC anchor	SRNC relocation	Path switch (MME, S-GW)	Path switch (AMF, SMF, UPF)
Idle area	LA (RA for PS)	LA + RA	TA list	TA list + RNA

Key takeaways

- Idle mobility is UE-controlled (reselection, area updates); connected mobility is network-controlled and mobile-assisted (measurement reports, handover).
- Area size trades update cost against paging cost: $N^* = (2c_u v/(\pi a c_p \lambda_c))^{2/3}$; TA lists let the network tailor it per UE.
- RSRP measures coverage, RSRQ $= N$ RSRP/RSSI adds load and interference.
- A3: $M_n - M_p > \text{Off} + \text{Hys}$ (plus offsets) for T_{TTT}; the handover point moves to $x = Dr/(1+r)$, $r = 10^{H/(10n)}$.
- With 8 dB shadowing a 3 dB margin is crossed by chance 40% of the time at the border: TTT and filtering are essential.
- X2 handover: report → request/ack → command → forwarding → RACH → complete → path switch → release.
- Too-late and ping-pong pull parameters in opposite directions; MRO, CHO and DAPS address the dilemma.

12 Exercises

Exercise 1 • Level 1 – conceptual • Who decides?

For each of the following, state whether the UE or the network takes the decision and on which information: (a) LTE cell reselection; (b) LTE handover; (c) WCDMA active-set update; (d) 5G conditional handover execution.

Exercise 2 • Level 1 – conceptual • Two update procedures

Explain why a GPRS-capable GSM phone could perform both a location area update and a routing area update at the same border, and how LTE avoided this duplication.

Exercise 3 • Level 2 – numerical • Offsets

In Example 1 the operator sets a cell-individual offset $O_{cn} = +2\,\text{dB}$ for the neighbour (to off-load the serving cell). Where is the A3 condition now met? Which failure type becomes more likely?

Exercise 4 • Level 2 – numerical • Filtering

Measurements are taken every 40 ms with independent errors of standard deviation 4 dB. Using Eq. (8), compute the standard deviation after filtering for $K = 4$ and $K = 8$, and the filter's time constant (number of samples for the impulse response to fall to $1/\text{e}$).

Exercise 5 • Level 3 – analytical • Optimal area with two classes

A tracking area serves 70% pedestrians ($v = 1.5\,\text{m/s}$, 1 call/hour) and 30% vehicles ($v = 15\,\text{m/s}$, 3 calls/hour). Using Eq. (9) with the other parameters of Example 3, find the single N that minimises the average cost, and the optimal N if each class had its own TA list. How much does per-class tailoring save?

Exercise 6 • Level 3 – analytical • Too late

Assume RLF occurs if the serving RSRP falls 8 dB below the neighbour before the handover command is received, and that preparation plus command delivery take 80 ms after the report. For the geometry of Example 1 without shadowing, find the maximum speed for which no too-late failure occurs with $H = 3\,\text{dB}$ and $T_{\text{TTT}} = 320\,\text{ms}$. Discuss which parameter you would change for a high-speed train line.

Solutions

Solution to Exercise 1

(a) UE, using broadcast parameters (Q_{rxlevmin}, Q_{hyst}, priorities) and its own measurements. (b) Source eNB, using UE measurement reports and its own load/admission knowledge. (c) RNC, using UE event reports (1A/1B/1C). (d) The UE executes, but only towards targets prepared by the network and when the network-configured condition holds – a shared decision.

Solution to Exercise 2

LA and RA belonged to two separate core domains (MSC/VLR and SGSN) with separate procedures. Unless the optional Gs interface allowed the SGSN to update the MSC on the phone's behalf (combined updates), the phone signalled twice. LTE has a single packet core with one mobility entity (MME) and one area concept (TA); CS-related registration, where needed for CS fallback, is performed by the MME over SGs.

Solution to Exercise 3

The condition becomes $M_n + 2 - M_p > 3$, i.e. $H_{\text{eff}} = 1\,\text{dB}$: $r = 10^{1/35} = 1.068$, $x_{\text{HO}} = 1000 \times 1.068/2.068 = 516\,\text{m}$. Handovers happen earlier; ping-pong and too-early failures become more likely, too-late less likely.

Solution to Exercise 4

Variance factor $a/(2-a)$. $K=4$: $a=0.5$, factor $1/3$, $\sigma_F = 4/\sqrt{3} = 2.31$ dB. $K=8$: $a=0.25$, factor $1/7$, $\sigma_F = 1.51$ dB. Impulse response $(1-a)^k$ falls to $1/\mathrm{e}$ after $k = -1/\ln(1-a)$: 1.44 samples (58 ms) for $K=4$, 3.48 samples (139 ms) for $K=8$. Better smoothing costs reaction time.

Solution to Exercise 5

Per-user cost $C_i(N) = 4c_\mathrm{u}v_i/(\pi a\sqrt{N}) + c_\mathrm{p}\lambda_i N$. The average has the same form with $\bar{v} = 0.7\times 1.5 + 0.3\times 15 = 5.55$ m/s and $\bar{\lambda} = 0.7\times 1 + 0.3\times 3 = 1.6$ calls/h, so $N^* = (2\times 5.55/(\pi\times 500\times 0.1\times 1.6/3600))^{2/3} = 29.3$, cost ≈ 14.1 units per user-hour. Separately: pedestrians $N^* = 16.8$ (cost 5.0), vehicles $N^* = 37.5$ (cost 33.7), average $0.7\times 5.0 + 0.3\times 33.7 = 13.6$. Tailoring saves only about 3% here, because the cost curve is flat near its optimum; it pays off more when the classes differ much more (e.g. stationary IoT devices versus trains). (The script `scripts/validate_examples.py` reproduces these values.)

Solution to Exercise 6

The RLF point is where $M_\mathrm{n} - M_\mathrm{p} = 8$ dB: $r = 10^{8/35} = 1.693$, $x_\mathrm{RLF} = 1000\times 1.693/2.693 = 629$ m. The report is triggered at 549 m and the command arrives after $T_\mathrm{TTT} + 80$ ms $= 0.40$ s. The UE must not travel more than $629 - 549 = 80$ m in that time: $v_\mathrm{max} = 80/0.40 = 200$ m/s. Without shadowing there is ample margin; with shadowing, fading of the serving cell can cut this distance drastically. For a train line, one would reduce TTT (speed-dependent scaling), lower the offset, deploy cells along the track with larger overlap, or use conditional handover.

13 Further reading

LTE mobility is described in [1], idle-mode procedures in [3] and the measurement events in [2]; GSM/UMTS handover in [4]; NR mobility, CHO and DAPS in [5]. Akyildiz et al. [6] survey location management and the fluid-flow model. Sesia et al. [8] and Holma and Toskala [7] discuss handover parameter tuning in LTE.

References

[1] 3GPP, *Evolved universal terrestrial radio access (E-UTRA) and evolved universal terrestrial radio access network (E-UTRAN); overall description; stage 2*, TS 36.300, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36300.htm.

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[3] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); user equipment (UE) procedures in idle mode*, TS 36.304, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36304.htm.

[4] 3GPP, *Handover procedures*, TS 23.009, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23009.htm.

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[6] I. F. Akyildiz, J. McNair, J. S. M. Ho, H. Uzunalioğlu, and W. Wang, "Mobility management in next-generation wireless systems", *Proceedings of the IEEE*, vol. 87, no. 8, pp. 1347–1384, 1999.

[7] H. Holma and A. Toskala, *LTE for UMTS: Evolution to LTE-Advanced*, 2nd ed. Wiley, 2011.

[8] S. Sesia, I. Toufik, and M. Baker, *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd ed. Wiley, 2011.